

ECON 1550: International Finance

# Fixed Exchange Rates and Foreign Exchange Intervention

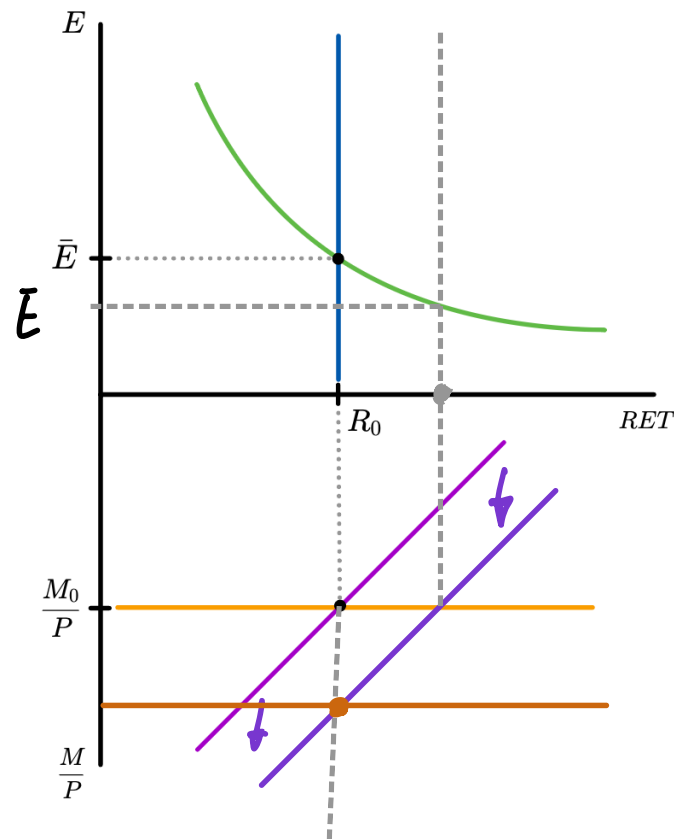
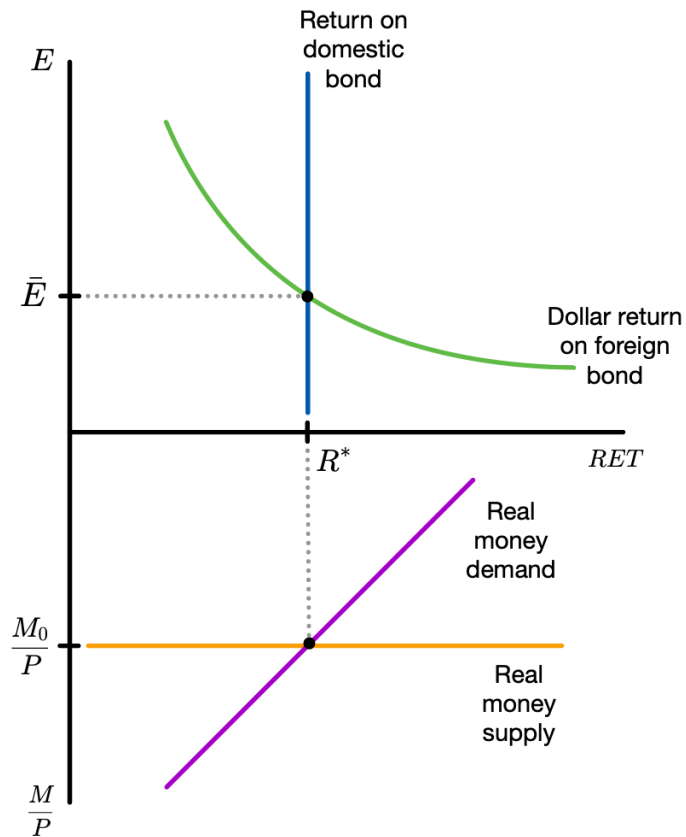
# How Central Banks Fix Exchange Rates

- Want to have a fixed exchange rate  $E = \bar{E}$
- Assume  $E = \bar{E}$  forever: the fixed exchange rate regime is said to be credible
- Then  $E^e = E = \bar{E}$
- Central bank changes money supply so that  $E = \bar{E}$  at all times
- UIP with  $E^e = E$  implies  $R = R^*$
- In a fixed exchange rate regime, a depreciation is called a devaluation and an appreciation a revaluation

# Response to Changes in $Y$

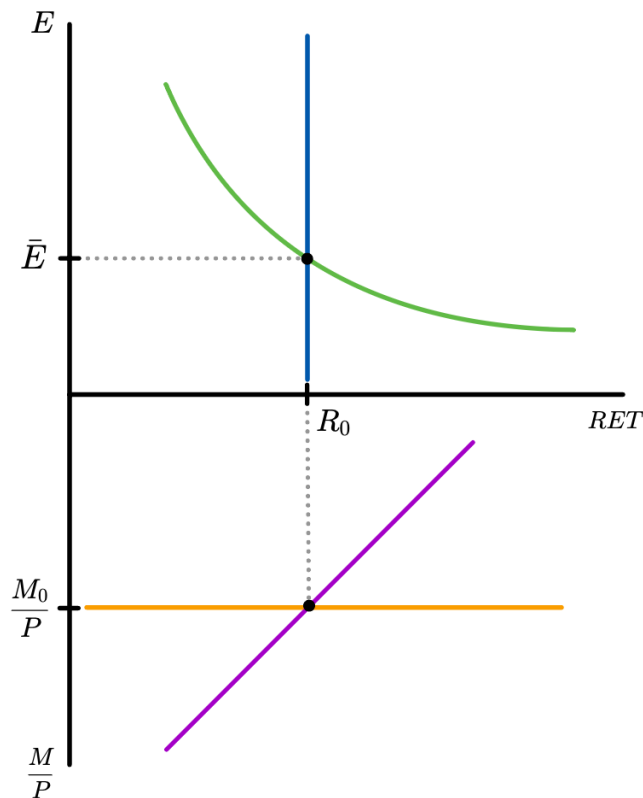
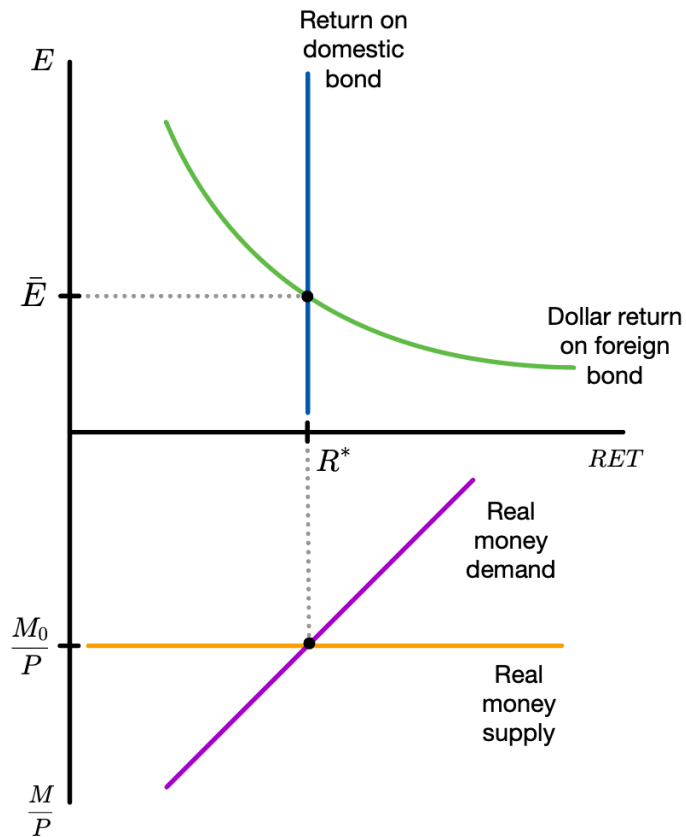
$Y \uparrow$

$Y^*$  UNCHANGED



# Response to Changes in $R^*$

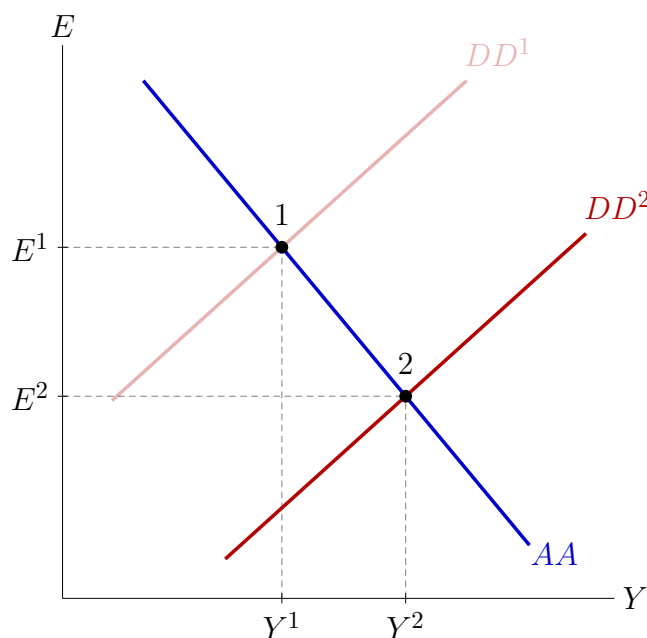
LEFT AS  
EXERCISE :)



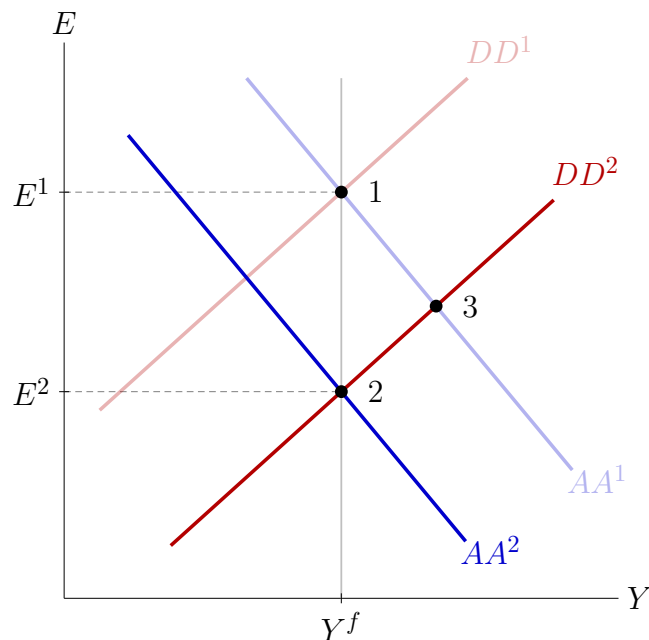
# Fiscal Stimulus

Floating exchange rate

Temporary

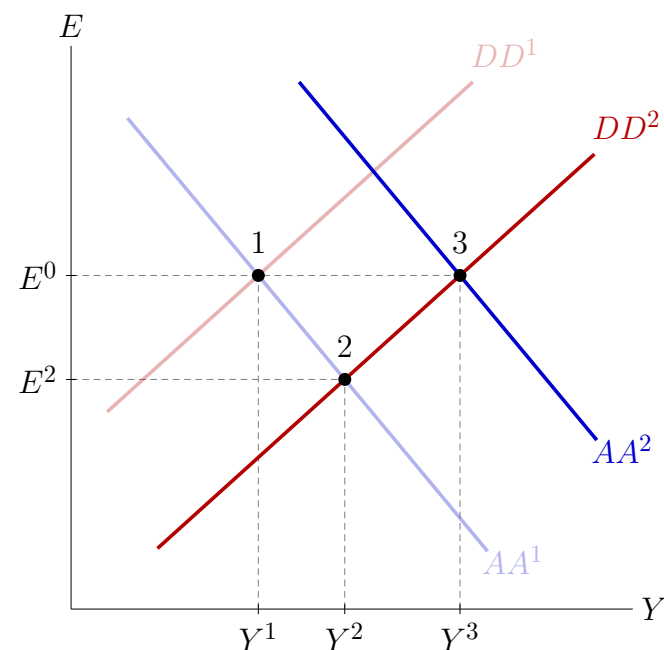


Permanent



Fixed exchange rate

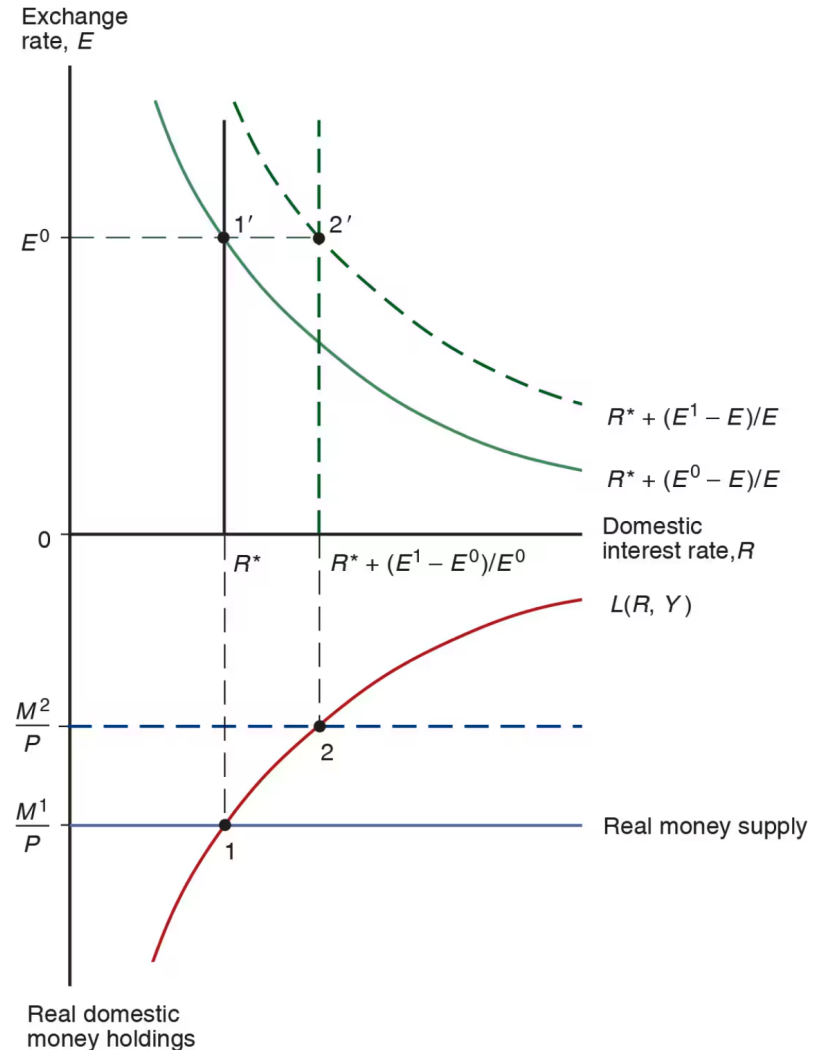
Temporary or permanent



# Balance of Payments Crises

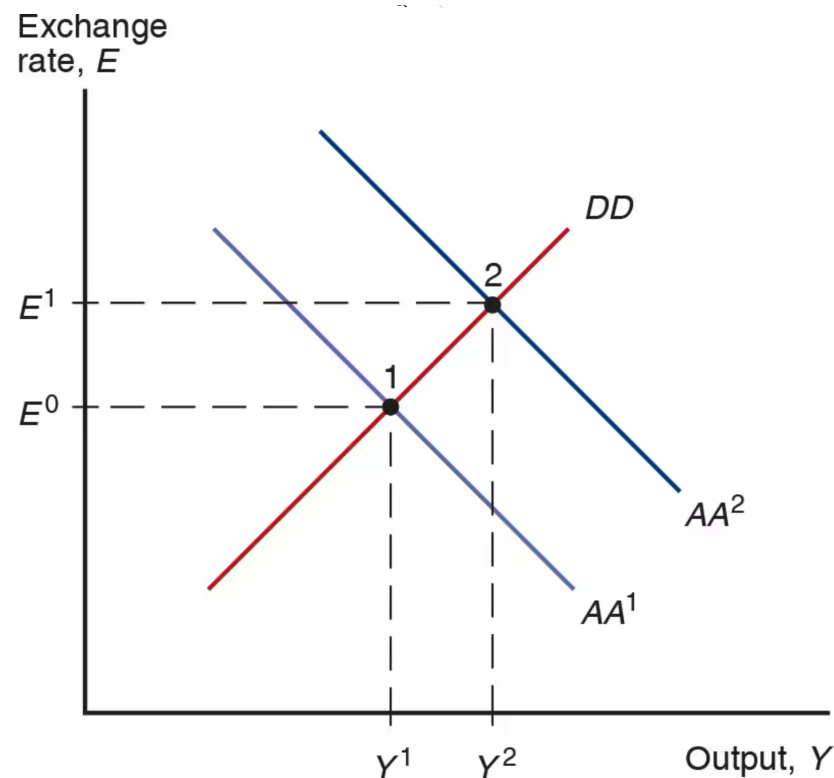
- Want to have a fixed exchange rate  $E = \bar{E}$
- If  $E^e = E = \bar{E}$  forever, the fixed exchange rate regime is credible
- Imagine  $E^e$  suddenly increases
- Without any government intervention,  $E^e < \bar{E}$

- To maintain  $E = \bar{E}$ , money supply must decline from  $M^1$  to  $M^2$
- The domestic interest rate must increase to  $R^* + E^1/E^0 - 1$
- As money supply declines,  $R$  is always below the dollar return on the foreign bond
- Investors flee the domestic currency and domestic bond: a **capital flight**



# Devaluation

- If central bank runs out reserves, or expects it will run out in the future, there will be a devaluation
- Fixed exchange rate must be modified or abandoned



1 PESO = 1 DOLLAR

FULLY CREDIBLE  $E = \bar{E}$ ? INSTITUTIONS  
MISTRUST.

END : DEVALUATION      3 PESOS = 1 USD

(1) 1 PESO = 1 USD  $\rightarrow$  APPRECIATION

(2) EX  $\downarrow$  IM  $\uparrow$   $Y \downarrow$   $u \uparrow$

(  $Y \downarrow$  PHILLIPS CURVE  $\pi \downarrow$  )

(3)  $G \uparrow$  DEFICITS GO UP

(4) TEQUILA & ASIAN CRISES

(5) BRAZIL (BIGGEST TRADING  
PARTNER) DEVALUES EX ↓

(6) G↑ BECAUSE Y↑ M↑

(7) HIGH GOV DEBT SOLVED BY:

↳ ABANDON  $E = \bar{E}$

↳ DEFAULT

↳ HIGH  $\pi \rightarrow$  ABANDON  $E = \bar{E}$

(8) PLOTS!

(9) GOV. COLLAPSE

↳ UNSTABLE GOV.

↳ DEVALUATION

↳ DEFAULT

(10) NEXT 3 YEARS WERE AMAZING  
↳ NO DEBT SERVICE  
↳ SUPER COMPETITIVE EX  
↳ GRAIN PRICES WORLDWIDE  
WENT UP.

⋮

(N) 10 YEARS LATER: DEFAULT  
AGAIN